Chaalel Omar Ziyad

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Objective

Aspiring AI & Software Engineer seeking knowledge and experience in everything theoretical & practical Computer Science has to offer while contributing to impactful, real-world projects.

Education

State Engineer in AI & Data Science Expected Graduation: Sep 2027
National Higher School of Artificial Intelligence (ENSIA), Algiers, Algeria.
Relevant Coursework: Intro to AI, Machine Learning, Data Mining, Algorithms & Data Structures, Operating Systems, Information Systems & Databases, Networks & Security, Reinforcement Learning.

Professional Experience

Data Science & AI Intern SLB Algeria (formerly Schlumberger)
Facies Classification using Well Logs September 2025

- Developed a machine learning model for facies classification from well log data, achieving 75% accuracy through systematic feature engineering and model optimization.
- Applied data imputation techniques to handle missing values in well log datasets, improving model robustness and predictive performance.
- Gained hands-on experience with Dataiku platform for end-to-end ML pipeline development.
- Acquired comprehensive understanding of SLB's main divisions and their operational workflows in the oil and gas industry.

Projects

Business Performance Analytics Dashboard May 2026

- Built an end-to-end analytics pipeline using **Python**, **SQL**, and **Power BI** to analyze retail sales, customer behavior, and product profitability from a large-scale transactional dataset.
- Developed interactive executive dashboards tracking key business KPIs, including revenue, profit margins, customer lifetime value (CLV), RFM segmentation, and regional sales performance.
- Implemented predictive analytics for customer churn and sales forecasting, alongside market basket analysis to identify cross-selling opportunities and support data-driven business decisions.

QoS-Aware Reinforcement Learning for SDN Routing May 2026

- Designed and implemented the **Dijkstra shortest-path baseline** and a **tabular Q-learning routing agent** for adaptive path selection in Software-Defined Networks (SDNs), enabling comparison between heuristic and learning-based routing strategies.
- Developed the complete SDN experimentation framework using **Ryu**, **Mininet**, and **OpenFlow**, including controller logic, flow installation, and deployment of Fat-Tree and Abilene network topologies.
- Built an automated traffic generation and QoS monitoring pipeline using **iPerf3** and **ping** to collect throughput, RTT, jitter, and packet-loss metrics for reinforcement learning feedback and evaluation.
- Evaluated reinforcement learning against Dijkstra and ECMP routing across multiple network topologies, demonstrating the advantages of adaptive congestion-aware routing under dynamic traffic conditions.

NABA2: Multi-Class Misinformation Detection Corpus for Algerian Darija 2025

- Co-led a 6-person research team to build NABA2, the largest multi-class Algerian Darija misinformation corpus to date. 22,391 annotated instances across 5 classes (Real, False, Misleading, Non-news, Satire), collected from Facebook, YouTube, Instagram, TikTok, and news platforms.
- Designed a hybrid data pipeline combining organic social media collection with LLM-assisted augmentation (Claude Sonnet 4.5, Gemini 3 Thinking) to offset class and dialect underrepresentation.
- Built a structured multi-annotator labeling protocol with priority-based conflict adjudication across 6 annotators; validated reliability at Fleiss' Kappa = 0.73 (substantial agreement).
- Benchmarked 8 models across classical ML, neural, and transformer architectures (BiLSTM, DziriBERT, mBERT, XLM-RoBERTa base/large); best model achieved 0.682 macro F1, a 5.4% improvement over the strongest classical baseline.

DDI Prediction with BioMedLM (RAG-based) March 2025 – May 2025

- Developed a Retrieval-Augmented Generation (RAG) system utilizing BioMedLM as main LLM to provide drug-drug interaction insights from biomedical texts.
- Aggregated data from three distinct medical sources (PubMed abstracts, DrugBank) to construct a unified knowledge base for retrieval.
- Contributed to improving system performance by ensuring the consistency of domain-specific input data.

Djezzy Dual-SIM Owners Detection January 2025

- Developed a graph-based dual-SIM detection framework over a telecom call network containing approximately **1M subscribers** and **2M directed interactions**, leveraging **NetworkX** for large-scale Social Network Analysis.

- Engineered behavioral similarity features using **Jaccard similarity**, common neighbors, interaction-frequency filtering, and edge exclusion to identify subscriber pairs exhibiting dual-SIM communication patterns.
- Designed a probabilistic scoring model combining graph-derived signals with CRM attributes (name, location, and age similarity) to estimate the likelihood of two SIM cards belonging to the same user.
- Built an end-to-end data mining pipeline including preprocessing, graph analytics, feature engineering, and visualization to support telecom customer behavior analysis and fraud detection.

Skills

- **AI & Machine Learning:** Python, NumPy, Pandas, Scikit-learn, TensorFlow, Reinforcement Learning Frameworks.
- **NLP:** Transformers, RAG pipelines, LLM-based annotation, IAA metrics, Hugging Face.
- **Computer Vision:** OpenCV, image processing and feature extraction.
- **Big Data & BI:** Hadoop (MapReduce, HDFS, YARN), Parallel Computing, Tableau.
- **Frontend:** React, TailwindCSS, Flutter, ElectronJS.
- **Backend:** Node.js, Express, Flask, PHP.
- **Databases:** OracleDB, SQL, Relational Modeling.
- **Networks:** TCP/IP, OSI Model, DNS/DNSSEC, Routing, IPv4 Addressing & Subnetting, NAT, GNS3, PacketTracer, Wireshark; proficient in configuring and deploying small network infrastructure.
- **DevOps & Tools:** Git, Docker, Linux.